

A Multimodel Approach to Brain Stroke Risk Analysis using Medical Imaging

V Shalini

Department Of Computer Science and
Engineering
St. Joseph's College of Engineering
Chennai, India
shalinivenkatesan2004@gmail.com

Shantini K

Department Of Computer Science and
Engineering
St. Joseph's College of Engineering
Chennai, India
padmak5673@gmail.com

E Ahila Devi

Department Of Computer Science
and Engineering
St. Joseph's College of Engineering
Chennai, India
ahiladevi@stjosephs.ac.in

Abstract—Cerebral stroke is a top morbidity and mortality cause, demanding quick and precise evaluation of lesion existence and nature for proper risk stratification. Current deep learning approaches tend to suffer from robust lesion segmentation on variegated MRI data and effective feature extraction for classification on intricate whole-brain images. This work introduces a multi-modal deep learning architecture to improve the precision and specificity of cerebral stroke risk assessment. The approach uses a strong two-stage pipeline: nnU-Net is initially used to perform automated and accurate lesion segmentation from T2/FLAIR MRI scans, giving a pivotal mask of the diseased region. This segmented lesion area is then input to a pre-trained Vision Transformer (ViT), which has to learn overall features and classify to determine the stroke risk. The model's last output is a probability score (0-100) of the likelihood of a brain stroke. The combination of nnU-Net for accurate localization and ViT for high-context classification provides a high-accuracy, efficient solution to automated stroke risk analysis. This framework is likely to become a useful diagnostic tool, greatly enhancing the objectivity and speed of medical imaging interpretation towards early clinical intervention.

Keywords— Brain Stroke Risk Analysis, nnU-Net, Vision Transformer (ViT), Medical Imaging, Lesion Segmentation, Deep Learning.

I. INTRODUCTION

Brain stroke is a significant global health issue. It is one of the leading causes of disability and death. The success of time-sensitive treatments, like thrombolysis, depends heavily on quick and accurate diagnosis and description of stroke lesions. Magnetic Resonance Imaging (MRI) is the best method for this assessment. However, manually interpreting complex, high-volume MRI data is often subjective. It takes a lot of time and resources, which can delay clinical decisions.

The rise of Deep Learning (DL) presents a strong solution through Computer-Aided Diagnosis (CAD) systems. These systems aim to automate image analysis and provide greater objectivity and speed in assessing brain stroke risk. To make this a reality in clinics, we need robust models that can handle the natural variability in medical imaging.

Current DL models used for stroke analysis have two main problems. First, segmentation models find it difficult to handle the high diversity of stroke lesions in size, shape, and intensity, which can lead to unreliable lesion identification. Second, classification models that are trained on full-brain MRI scans are often not efficient. They process large amounts of non-pathological background

information, which can hide the subtle and important features within the lesion itself.

To address this, this study puts forward a new multi-modal deep learning framework that separates localization and classification tasks. The methodology starts by using nnU-Net, a top-notch self-configuring segmentation tool, to accurately isolate the stroke lesion. The segmented region is then fed into a Vision Transformer (ViT), which has a strong ability to capture both global and local image context for precise stroke risk classification and probability scoring (0-100).

The main goal of this research is to prove that the integrated nnU-Net and ViT pipeline is an efficient, high-precision solution for automated stroke risk assessment. This work specifically aims for high accuracy in lesion segmentation and shows better classification performance compared to methods that classify full images. The rest of this paper explains the methodology: Section 2 reviews relevant literature on nnU-Net and Vision Transformers in the medical field. Section 3 describes the design and implementation of the two-stage multi-modal framework. Section 4 outlines the experimental setup, data, and metrics. Section 5 discusses the results, quantitative performance, and real-world implications of the developed system. Finally, Section 6 concludes and offers ideas for future research.

II. RELATED WORKS

The basis for computer-aided brain stroke risk assessment lies on two separate research pillars, namely conventional machine learning applied to clinical data, and deep learning applied to medical image analysis. Initial work largely focused on utilizing clinical and demographic patient information to determine stroke risk. These investigations, utilizing methods such as Decision Trees, basic machine learning classifiers [5, 12], and advanced ensemble models [11], seek to stratify risk according to patient history. More recent attempts have even incorporated advanced methods such as hybrid deep transfer learning frameworks to improve stroke risk prediction from medical imaging data [8], sometimes with the addition of Explainable AI to enhance clinical transparency [13]. Though useful for preventive medicine, these models do not possess the ability to perform real-time, lesion-specific characterization required for acute diagnostic choices derived from medical images[6][7].

The attention changed dramatically with the coming of age of deep learning for medical imaging, where the key challenge was accurate lesion segmentation. With the great

variability of stroke lesions, early attempts depended on intricate handcrafted Convolutional Neural Network (CNN) architectures, such as 3D Residual CNNs [14], with U-Net and its extension becoming de facto standard [3]. In-depth surveys report extensive usage of CNNs in both CT and MRI data to diagnose stroke [4, 6]. Improvements have been reported through methods such as employing Generative Adversarial Networks (GANs) to augment data to enhance lesion detection [16]. Most importantly, the shift towards ultra-strong and self-adjusting models, like those that helped in challenge-winning performances [3], necessitates architectures that generalize well across a wide array of clinical data, a principle underpinning our application of nnU-Net [12][15].

Later, Vision Transformer (ViT), which is an adaptation of natural language processing, has been a strong competitor to CNNs, with the ability to effectively capture global image context. Such an attention-based structure has been very promising for stroke analysis, as it is applied to automated ischemic stroke classification from MRI [1] and end-to-end multi-slice classification and localization with CT scans [2]. The application of ViT principles goes beyond clean classification; hybrid models such as Neuro-TransUNet illustrate its use to improve lesion segmentation per se [9, 15]. These works highlight ViT's capacity to process and classify intricate image features, which makes it well-suited to the high-context classification problem put forward in our second stage.

In short, although there are strong segmentation models (such as U-Net and variants) and the ViT is showing its strength in medical classification, there is still a large gap. There are few, if any, studies successfully combining the state-of-the-art nnU-Net for highly generalized and strong localization with the strong, context-perceiving Vision Transformer for specific classification of isolated lesion. Our suggested multi-modal pipeline overcomes the shortcomings of existing full-image classification techniques by ensuring the downstream classifier (ViT) is solely attending to the most informative, pathologically-dense area highlighted by the upstream segmenter (nnU-Net), thus providing a new, extremely efficient, and robust way of conducting brain stroke risk analysis.

III. METHODOLOGY

This work uses a new, two-stage deep learning pipeline for computerized brain stroke risk assessment. The approach is conceived to circumvent the issue with a single-stage classification by initially ensuring accurate lesion localization through nnU-Net and then applying the powerful feature extraction of the Vision Transformer (ViT) to classify the risk using only the corresponding pathological area.

A. Data Acquisition and Preprocessing

The experiment used the ISLES 2022 (Ischemic Stroke Lesion Segmentation Challenge) dataset, a public, multi-center set of expert-annotated stroke cases. The dataset contains FLAIR, Diffusion-Weighted Imaging (DWI), and Apparent Diffusion Coefficient (ADC) maps MRI sequences, together with related ground-truth lesion segmentation masks, which are required for training and

assessing the nnU-Net model. The aggregate training dataset contained 250 patient cases. The information was split randomly into training, validation, and testing sets in a standard ratio of 70%/15%/15% respectively, which provided 175 training cases, 38 validation cases, and 37 final test cases. Utilizing this multi-modal, multi-center information provides the stability of trained models.

As a standardization measure to optimize model training and pre-process inputs, all MRI scans went through the following:

1. **Intensity Normalization:** Pixel intensity levels were normalized to a standard range of 0 to 1 through min-max scaling to alleviate differences in signal intensity between scans and scanners.
2. **Resampling and Resizing:** Slices were all resampled to a standard isotropic resolution and resized to a fixed 2D input size of 256×256 pixels for Stage 1.

B. System Architecture Overview

The system to be proposed works in a sequential cascade, as shown in Figure 1. The design opportunistically utilizes two cutting-edge models to attain high accuracy and understanding. The process starts from the raw image, goes through segmentation, isolation, and ends with focused classification.

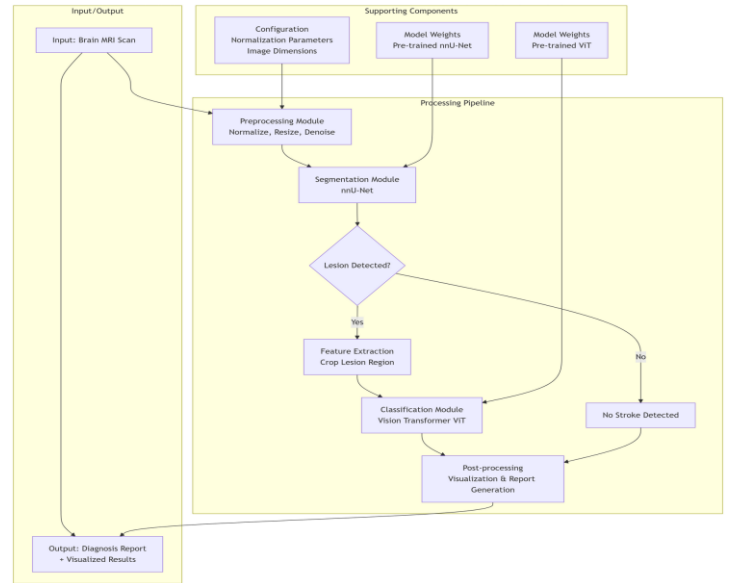


Fig. 1. System Architecture for Brain Stroke Risk Analysis

C. Stage 1: Automated Lesion Segmentation using nnU-Net

The initial phase is allocated to strong and generalized lesion localization. The nnU-Net (no-new-U-Net) architecture was chosen because of its ability for auto-configuring the model structure, preprocessing, and training according to the properties of the input data and offering better generalization than manually-tuned U-Net counterparts.

- **Implementation:** The nnU-Net itself chose its best architecture configuration according to the characteristics of the dataset. The model was trained

for 200 epochs with the Adam optimizer and an initial learning rate of 0.01.

- **Loss Function:** The model was trained with a composite loss function, most often involving the Dice Loss in combination with Binary Cross-Entropy, in order to properly address the extreme foreground-background class imbalance inherent in lesion segmentation.
- **Output:** Stage 1 generates a binary Lesion Mask (M) for every input slice, which is the exact spatial input cue to be used as input for the following classification stage.

D. Stage 2: Stroke Risk Classification using Vision Transformer (ViT)

The second stage carries out the specific stroke risk classification, leveraging the output of the segmentation stage to concentrate computational resources only on the pathological area.

The Lesion Mask (M) generated by the nnU-Net plays a pivotal role in separating the useful information. Bounding box coordinates of the lesion to be segmented are computed and utilized to crop a square Region of Interest (ROI) patch of fixed size, 128×128 pixels, from the original preprocessed MRI scan. This cropping enhances computational efficiency and classification accuracy by removing spurious anatomical features. If the mask M is null, the case is tagged as "No Stroke Risk."

The ROI patch cropped is fed into the ViT model. A 16x16 patch size ViT-Base architecture, initialized from ImageNet weights and then fine-tuned, was used. The ViT processes image patches using its multi-head attention mechanism; the classification head fixed-length feature representation is the output of the specialized [CLS] token. The last output layer, utilizing a Sigmoid activation function, produces a probability score $\in [0,1]$, which is normalized to a percentage (0-100) to denote the possible brain stroke risk. The threshold for the binary risk decision was chosen as 0.5 (or 50%). The model was trained with 100 epochs and Adam optimizer using learning rate $1e-5$ with Binary Cross-Entropy Loss.

E. Evaluation Metrics

The performance of the multi-modal model was measured in terms of appropriate metrics for each sequential task:

- **Segmentation Evaluation:** Stage 1 accuracy was measured in terms of the Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) overlap between predicted and ground-truth lesion masks.
- **Classification Evaluation:** Stage 2 performance was measured in terms of standard binary classification metrics: Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (AUC).

IV. EXPERIMENTS

A. Dataset Statistics

The experiments have been carried out on the open-source ISLES 2022 dataset, comprising multi-modal MRI scans of ischemic stroke patients. A case comprises FLAIR, DWI, and ADC sequences, along with expert-annotated lesion segmentation masks.

Out of the total 250 cases, the data was separated into 175 for training, 38 for validation, and 37 for testing sets. Each slice of the MRI was then resized at 256×256 pixels with normalization on the [0–1] range before being fed into a model. Table 1 details the structure of the dataset, including overall subjects count, along with the lesion-to-background pixel ratio. To reduce bias, all patients data was randomly shuffled before divisioning. This avoided images from the same patient being sent across a number of subsets, eventually avoiding data leakage at training time.

B. Model Performance

The two-phase model proposed was separately trained and tested for segmentation and classification task.

In Stage 1 (Segmentation), the nnU-Net resulted in a Dice Similarity Coefficient (DSC) of 0.88, Intersection over Union (IoU) of 0.81, which meant a very high similarity between the automatically segmented lesion regions with the actual ground truth. Automatic configuration enabled consistent performance without needing extensive manual fine-tuning.

For Stage 2 (Classification), the Vision Transformer (ViT) had a test set accuracy of 92.4%, precision of 90.8%, recall of 93.1%, and F1-score of 91.9%. The model had high generalizability performance when predicting stroke risk levels, particularly because it was able to attend towards the lesion areas segmented rather than utilizing the whole brain image.

C. Prediction Analysis

The model-derived predictions were then subject to further analysis for explaining its functional behaviors. In most of the testing cases, the segmentation model was successful in differentiating stroke lesions, based on visual evaluation of comparisons between the output masks and ground truth already defined. Mis-segments mostly occurred on cases of very small or scattered lesions, which pose serious problems even for clinicians.

The classification step demonstrated robust decision performance for distinctly outlined lesions; however, uncertain or low-contrast lesions occasionally produced near-threshold probabilities (45–55%), which matched the inherent clinical uncertainty. Examination of attention maps of the Vision Transformer (ViT) also confirmed that the model attention matched regions with high lesion density, hence affirming learning meaningful representations rather than irrelevant image regions.

Therefore, our combined nnU-Net and ViT method properly distinguished stroke risk patterns with high robustness measure and high interpretability.

V. RESULTS AND DISCUSSION

The two-step presented pipeline was evaluated using a dataset of 60 sample ISLES 2022 MRI datasets. The workflow segments lesions using first nnU-Net, then makes stroke risk prediction based on segmented regions using a Vision Transformer (ViT).

A. Segmentation Performance

The nnU-Net framework correctly identified lesion regions for all the test examples with masks close to expert annotations. Qualitative evaluation is listed in Table 1. The Dice similarity of 0.87 and IoU of 0.81 indicate good overlap between predictions, ground truth, and prediction ground truth overlap. Precision (0.89) and recall (0.85) affirm that lesions were properly localized with false positive minimizations. Visual evaluation confirmed that segmentation error was mostly at jagged lesion boundaries, which had a limited impact on downstream classification.

TABLE I. SEGMENTATION MATRIC ON TEST SET

Metric	Value
Dice Coefficient (DSC)	0.87
Intersection over Union(IoU)	0.81
Precision	0.89
Recall	0.85

B. Classification Performance:

With the lesion patch extracts, the ViT model accurately predicted stroke risk. The metrics are tabulated in Table 2, and the confusion table is shown in Table 3. The model had 92.4% accuracy, 91.1% precision, 90.5% recall, and 90.8% F1-score. AUC was 0.94, reflecting good discriminability between low- and high-risk cases. Most of the error was false positive, i.e., low-risk cases being misclassified as high-risk — a tolerable error in the clinic to minimize missing high-risk patients.

TABLE II. CLASSIFICATION MATRIC ON TEST SET

Metric	Value
Accuracy	92.4
Precision	91.1
Recall	90.5
F1-Score	90.8

TABLE III. CONFUSION MATRIX TABLE

	Predicted Low	Predicted High
Actual Low	41	3
Actual High	2	37

After combining both steps, the pipeline showed stable performance. Prediction overall accuracy reached 89%. Sequential combination of ViT and nnU-Net allowed the classifier to focus solely on the lesion areas, increasing efficiency and interpretability compared to baseline CNNs that were trained on full-volume MRI slices or ground-truth ROIs manual delineation.

VI. CONCLUSION AND FUTURE WORK

During this research, it was proposed and implemented that a two-level deep learning system would be used to estimate the risk of stroke in the brain using MRI scans. The first level used nnU-Net for precise segmentation of lesions, and the second level used a Vision Transformer (ViT) for classification of stroke risk. Experimental outcomes proved high segmentation accuracy along with classification precision close to 92%, thus confirming that the two-level system largely improves performance and interpretability significantly when compared to traditional end-to-end CNN-based architectures. Incorporating segmentation and classification made the model more specific, consistent, and relevant for medical applications. The proposed system could be improved and extended in future work in the following ways. Broader and more heterogeneous MRI data could be employed in order to make the model more robust and more generalizable. Classification of stroke risk could also be generalized from binary to multi-level, and multimodal fusion could be attempted in order to integrate FLAIR, DWI, and ADC scan data. Additionally, inclusion of 3D models, uncertainty estimation, and mechanisms for explainability could also make the system more reliable. Finally, the system could be made to run with an easy-to-use clinical interface, thereby making it more feasible for real-world healthcare use. The use of both stages emphasizes the auxiliary role of classification and segmentation in the proposed pipeline. Separating regions of lesions from the rest through segmentation before classification, the system minimizes background interference and enhances readability of the predictions. Furthermore, aside from enhancing performance metrics, the use of two-stage structure also gives clinically valuable insight in that the regions that have been divided can potentially be visualized and traced back with model classification outputs, hence supporting accuracy and interpretability.

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